Sort K Sorted Array

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The problem

- Every element is at most k positions from its sorted position; sort with a size-(k+1) min-heap.

Intuition

- Every element is within k of its spot, so the minimum of the next k plus one items is always next in order.

Approach

- 1. Fill a min-heap with k plus one items
- 2. Pop the min into place
- 3. Push the next item
- 4. Drain the heap

Java solution

```java
public class SortKSortedArray {
    public static int[] sortKSortedArray(int[] a, int k) {
        PriorityQueue<Integer> pq = new PriorityQueue<>();
        int w = 0;
        for (int i = 0; i < a.length; i++) {
            pq.add(a[i]);
            if (pq.size() > k) a[w++] = pq.poll();
        }
        while (!pq.isEmpty()) a[w++] = pq.poll();
        return a;
    }
}
```

Complexity

- Time: O(n log k)
- Space: O(k)

Recap

- Every element is within k of its spot, so the minimum of the next k plus one items is always next in order.
- Complexity: O(n log k)